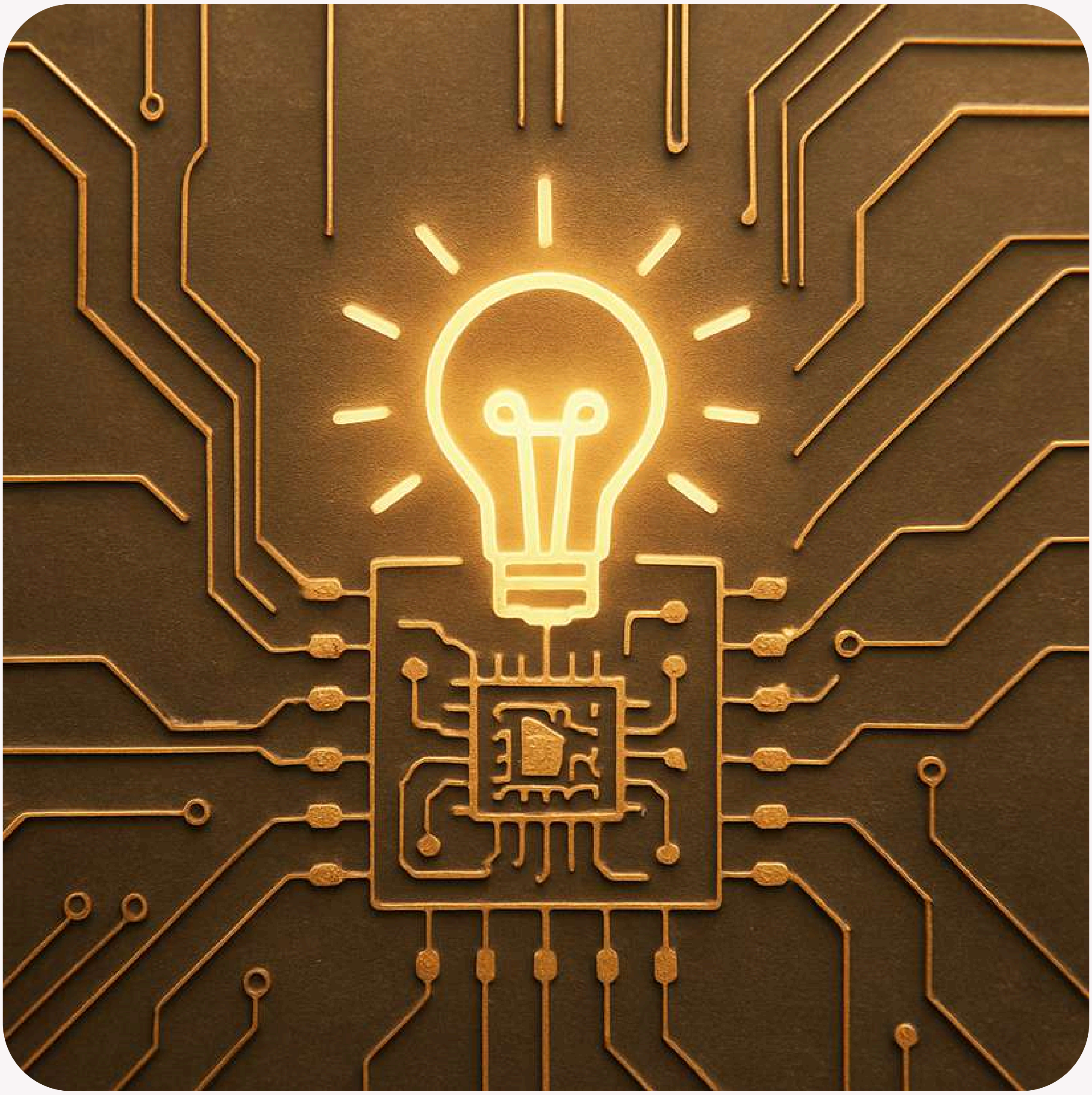
Annual Newsletter



2024 -2025

GREETINGS FROM DECCTEROUS IC DESIGN CLUB!

Humble Beginnings - *2024*

Welcome to the 2024 edition of our club newsletter! This year has been an exhilarating journey filled with knowledge, innovation, and collaboration. Our club has consistently strived to provide a dynamic platform for students passionate about IC design, fostering technical skills, research, and industry exposure. Through a series of workshops, competitions, and expert sessions, we have created a thriving community where ideas transform into reality.

With a vision to bridge the gap between academia and industry, DECCTEROUS IC DESIGN CLUB has continually empowered students by offering hands-on experiences and interactions with industry professionals. Our events have not only enhanced technical acumen but also cultivated creativity and teamwork among participants. As we reflect on an eventful year, we take pride in sharing the highlights of our journey in 2024. From hands-on workshops to industrial visits, our club has been at the forefront of innovation and learning. Here’s a glimpse of the remarkable events that shaped our year.

OUR MISSION

The mission of the Integrated Circuit Design Club is to cultivate an environment of experiential learning where members are empowered to acquire and enhance the skills necessary for a successful career in the core Electronics and Communication domain. Through hands-on projects, industry interactions, and collaborative initiatives, the club strives to nurture technical competence and innovation among its members.

OUR VISION

The Integrated Circuit Design Club envisions becoming a vibrant center of excellence in both analog and digital circuit design, providing innovative solutions for various applications. By fostering strong collaboration between industry and academia, the club aims to address the evolving needs of core industries, thereby bridging the gap and promoting technological advancements in the field.

PRINCIPAL'S MESSAGE

Dr. Rohini Nagapadma

Principal NIE

It brings me immense joy to release the Newsletter of the Deccterous IC Design Club. This club promises to bring out real motive innovation, technical expertise & collaborative learning among our students. IC design is at the heart of modern technology, powering devices that shape our daily lives & define the future. Through this club, we aim to provide a platform where students can explore the fascinating world of IC design, enhance their technical skills, & work on projects that make a tangible impact. I congratulate all the faculty coordinators & wish the club a great success.

ECE HOD'S MESSAGE

Dr. S Parameshwara

Associate Professor and HoD

It gives me great pleasure to witness the release of the newsletter of the IC Design Club "Deccterous."

This publication reflects the creativity, dedication and teamwork of our vibrant student community. Clubs play a vital role in nurturing talents beyond the classroom, and this newsletter stands as a testament to the enthusiasm and commitment of the students involved.

It is encouraging to see such thoughtful content,event highlights and achievements being documented and shared with pride.

May this newsletter inspire more students to participate, collaborate and grow together.

Wishing the IC Design Club "Deccterous" continued success in all its future endeavors.

FACULTY COORDINATORS-2024-'25

Dr. Salila Hegde
Associate Professor, *ECED*

Dr. Remya Jayachandran
Assistant Professor, *ECED*

OFFICE BEARERS-2024

Sujala B R
President

Sanjay Hanchinal
Vice-President

Disha L
Secretary

Shesha Krishna S
Treasurer

Preksha G M
Registration Head

Vaishnavi Purshottam
Registration Head

DhanyaShree
Registration Head

Thanushree M S
Event Organizers

Shrilakshmi
Event Organizers

OFFICE BEARERS-2024

Skandan Bharadwaj K
Publicity,Marketing,
Sponsorship

Satyadutt M S
Publicity,Marketing,
Sponsorship

Jayesh Gidwani
Technial Head

Nidishree
Technial Head

Adhrutha S M
Project & Industry Visit

Pranav B S
Project & Industry Visit

Bheema Kashyapa
Project & Industry Visit

Vishanth
Project & Industry Visit

Shraddha S A
IdeaLab Projects

OFFICE BEARERS-2024

Shivabasavesh A S
IdeaLab Projects

Mourya P
IdeaLab Projects

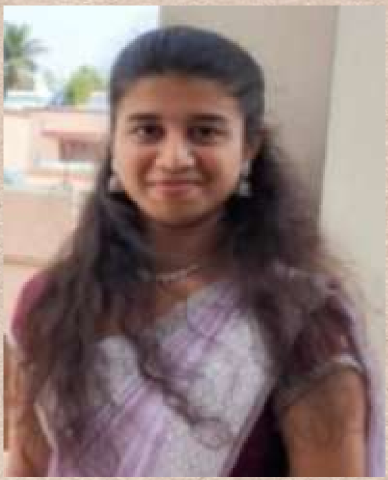
Parushuram M
Website & Social Mediia

Sagar Singh
Website & Social Media

Anagha
Website & Social Media

Harshitha B M
Website & Social Media

Varshini H R
Hospitality



Neha S
Hospitality

Hemashree
Hospitality

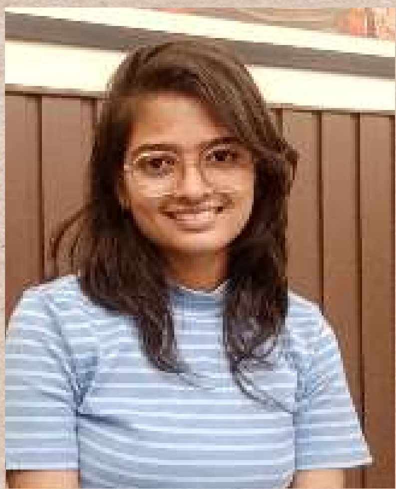
OFFICE BEARERS-2024

Prekshadeep
Design Creation

Shreyansh Shrivatsav
Design Creation

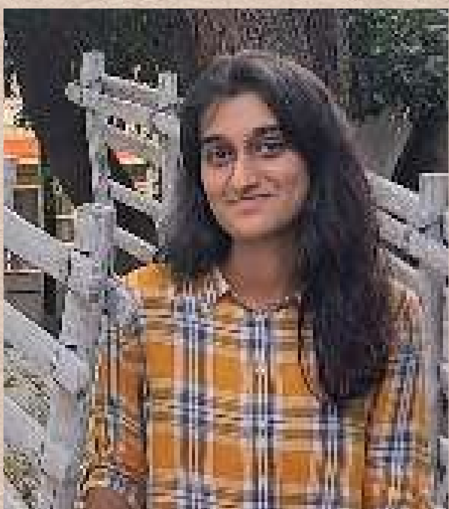
Pankaja
Coding & Programming

Mehul J
Coding & Programming



Aishwarya
Coding & Programming

Harsha B
Stage Committee



Sanjana R
Stage Committee

CHIEF PATRONS

Dr. Rohini Nagapadma
Principal, NIE Mysuru

Dr. Manjunath K C
Vice Principal, NIE Mysuru

ADVISORY COMMITTEE

Dr. Narasimha Kaulgud
Professor,
ECE, NIE Mysore

Dr. Parameshwara S
Associate Professor & Head,
ECE, NIE Mysore

INDUSTRY

Dr. Nisha Haridas
Senior Engineer,
Qualcomm

Dr. Supriya Unnikrishnan K
Principle Engineer,
Ignitarium

Mr. Vijayakrishnan K K
Senior Staff Engineer,
Infineon Technologies

Mr. Edet Bijoy K
Digital IC Design
Engineer,
KaikuTek, Taiwan

Haripriya Kesavan
NVE SSD Senior Product
Engineer

Dr.Arjun Hari M
Senior Facility Technologist-
Process Integration National
Nanofabrication Centre
CeNSE-IISc-Bangalore

Dr. Gayatri Parthasarathy
R&D, BEL

Dr. Deepak P M
CEWIT, IIT Madras

Dr. Shaeen Kalathil
Assistant Professor, EEE
Princess Nourah Bint University,
Saudia Arabia

Dr. Nithin Thomas Abraham
Principal Design Engineer
ULP CMOS-Analog Reference
Design Bengaluru
GlobalFoundries India.

NIE-ALUMNI

Mr. Charan M S
Analog Layout Engineer,
Texas Instruments,
Bangalore

Mr. Skanda Prasad
Site Reliability Engineer
Cisco Bangalore

Mr. Prabhanjana S K
Software Developer Engineer
Seimens Bangalore

Saikiran Babu Annangi
Data Science Master's Student
University of Washington

Mr. Santhosh H S
AI Engineer & Data
Scientist @ TCS-AI Lab

Mr. Aniruddh Adiga
Master's in Data Science
Texas A&M University

Ms. Sangeetha A Shayana
HVAC Engineer
Schneider Electric,
Bangalore

Mr. Pundlik Anant Nayak
Analog Layout Engineer,
Texas Instruments, Bangalore

Ms. Raksha A R
Analog Layout Engineer,
Texas Instruments,
Bangalore

Mr. Amith N Bharadwaj
R&D Engineer,
Tejas Networks Ltd.
Bangalore

Ms. Lekhana Gowlla
R&D
Mercedes Benz
Bangalore

Mr. Suhas Shastry H S
Associate Engineer
Morphing Machines,
Bangalore

Mr. Gurusatwik Bhatta N
Project Associate
IISc Bangalore

Mr. G Adarsh
Associate Support Engineer
Blue yonder India Pvt Ltd.
Bangalore

Mr. Darshan Babu
Senior HW Engineer
Samsung
Bangalore

Ms. Ananya Murali Kashyap
Masters student in Electrical Engineering ,
University of Pennsylvania

Events Conducted in 2024:

1) Inauguration ceremony

2)Innovation & Design Thinking Project Expo 2024

3)PCB Design and arm Microcontroller Hands-on Workshop

4)Career Guidance Program

5)Orientation program

6)Treasure Hunt

7)3D Printing Workshop

8)Hands-on Session on Analog CMOS Circuit

9)Quizmosis

10)Industrial Visit

11)Digital Poster Design

12)Analog Showdown

1)INAUGURATION CEREMONY

Date: July 6, 2024
Venue: Sir MV Hall, Golden Jubilee Complex
Organized by: IC Design Club, NIE
Participants: 127 students
The launch of the IC Design student club, DECCTEROUS, dedicated to innovation in analog and digital circuit design.

Guest Speakers & Industry Experts:

Alumni:

Mr. Charan M S (Texas Instruments),Mr. Prabhanjana S K (Siemens), Mr. Pundlik Anant Nayak (Texas Instruments), Miss Sangeetha Shayana (Schneider Electric)

Industry Experts:

Dr. Nisha Haridas (VLSI, Qualcomm), Dr. Supriya Unnikrishnan (VLSI, Ignitarium), Mr. Vijayakrishnan K K (Infineon), Mr. Edet Bijoy (Digital IC Design Engineer, KaikuTek, Taiwan)

Financial Support: ₹17,000 granted by the college to fund club activities.
Logo Reveal & Recognition: The unveiling of the club logo, with awards for the best logo design and naming contributions

1)INAUGURATION CEREMONY

The IC Club hosted its inaugural event, which featured a range of engaging activities designed to foster collaboration and innovation among students. The event commenced with an interactive quiz competition that involved both first- and second-year students, encouraging teamwork and enhancing their knowledge of integrated circuits.In addition to the quiz, the event included evaluations of innovative projects. First-year students presented their IDT (Innovative design thinking) projects, showcasing their creativity and technical skills. Meanwhile, second-year students participated in the evaluation of their analog circuit projects, demonstrating their advanced understanding of circuit design.

The winners of the quiz were Pankaja Patil from the first year and Varshini H R from the second year, both of whom achieved the highest scores.

M N Varshini and Anusha K G from A section, along with Thanushree MS and Skandan Bharadwaj from B section, were recognized for their outstanding performance in academics during the second year.

Harshitha BM from A section and Vikas from B section were awarded prizes for their exceptional work in logo design and naming for the club.

The launch of DECCTEROUS marked a major milestone for NIE Mysore’s ECE department. A strong foundation for future IC design projects and collaborations was established. With continued support from faculty and industry experts, the club is set to drive innovation in circuit design.

2) Innovation & Design thinking Project Exhibition

Innovation & Design thinking project Evaluation & Demo which is one of the courses for the ECE branch students. Students were able to emphasise the literature survey, define the problem, development of prototypes and testing, challenges, failures, feedback from evaluators technical communication skills, documentation, and presentation skills.

Prize winners:

The winners were selected based on both internal and external evaluation. Phase 1+Phase 2+ Final Evaluation marks were calculated. Two tracks were there- NICU track and Waste Management Track. The first prize was cash prize of Rs. 1250/- (Two teams), the Second Prize- of Rs. 750 /- (four teams) and Third Prize - Rs. 500/- (six teams). The funding was received from our department fund to IC Design Club to promote student Development programs.

TRACK 1: NICU

First Prize - Priya R, and team - Team No.2

Second Prize - Suyajnaa and team - Team No. 13

Third prize - Vishanth and team-Team No.4, Shrilakshmi and team -Team No.5

TRACK 2: WASTE MANAGEMENT

FIRST PRIZE - Neha S Kamal and team - team No.6

Second prize - Harsha B and team- Team No. 2,

Third Prize - Dhanyashree and team -Team no.3, Advaith & team -Team No.5, Vikas and Team -Team No. 3

For the second-year students on behalf of the club there was Analog Electronic Circuits Design Project.

List of prize winners-

From “A” section:

1st prize- Adrutha SM and team.

2nd prize- Ksheeraj S.K and team,Milana G and team.

3rd prize- Dhanraj.R and team, Ankitha A Renjal and team,Karthik MN and team,Deepu and team.

From “B” section:

1st prize-Shesha Krishna S and team.

2nd prize-Reefa Tahaniyat and team,Varshini H.R and team.

3rd prize-Vaishanavi T Patil and team,Rohan S Paraddi and team.

3)PCB DESIGN AND SENSOR INTERFACING TO ARM MICROCONTROLLER WORKSHOP

The PCB Design and Sensor Interfacing to ARM Microcontroller Workshop was successfully conducted on 6th & 7th September at the DSP Lab and ARM Lab, NIE, organized by DECCTEROUS - IC Design Club in collaboration with the NIE-AICTE Mechatronics Lab.

The workshop was guided by esteemed mentors Dr. Salila Hegde, Dr. Remya Jayachandran, Mr. Nithesh K A, Mrs. Pushpalatha H P,and Mr. Lokesha C, under the patronage of Dr. Rohini Nagapadma (Principal, NIE), Dr. S Ganesh Prasad (Vice Principal, NIE), and Dr. S Parameshwara (HoD, ECE).Participants gained valuable hands-on experience in PCB design and sensor interfacing with ARM-based microcontrollers, applying their knowledge to real-time applications. The interactive sessions enabled students to enhance their circuit design and implementation skills, all while working in state-of-the-art DSP and ARM laboratories.

The event was coordinated by students Adhrutha S M and Shesha Krishana, whose efforts were instrumental in the smooth execution of the workshop.

4)CAREER GUIDANCE PROGRAM

The IC Design Club successfully hosted a Career Guidance Program titled "Deep Dive into the Mindset of the Interviewer" on September 28, 2024, at MV Hall, GJB Block, NIE. Organized by the Department of ECE, the session provided students with key insights into acing technical round interviews. The event featured distinguished industry professionals and alumni, who shared their experiences and strategies for interview success. The program was coordinated by Mr. Remya Jayachandran (Assistant Professor, ECE Dept.) and Dr. Salila Hegde (Associate Professor, ECE Dept.), under the guidance of Dr. Parameshwara S (HoD, ECE Dept.). The experts were alumni of NIE who are working in core EC and IT industries.

They shared their experience and gudied students about how to prepare and face interviews. Attendees engaged in interactive discussions, gaining valuable knowledge on technical expectations, interviewer perspectives, and effective preparation techniques, making the session highly beneficial for students preparing for placements.

The logo was relaunched on that day – the logo was designed by Shreyansh and team.

<u>Winners of the club-organized events receive certificates and cash prizes.</u>

Career guidance Program conducted for fifth semester students . "DECCTEROUS"- IC Design Club of ECE Department NIE Mysore organized a Basic Placement Training Program for Fifth Semester ECE Students on 11/09/2024 at The National Institute of Engineering. The event, conducted by the IC Design Club, featured Mr. Abhimanyu Dwibedi, an esteemed alumnus of the 2024 pass out batch from CSE. Mr. Dwibedi, who is working as an Associate Software Developer at ZSCALER, Bangalore, shared his insights and expertise with the students

5) Orientation Program

The students from the 3rd and 5th semesters of the Electronics and Communication department engaged with the newly admitted first-year students of the ECE department. They introduced them to the IC Design Club, highlighting its activities and the advantages of becoming a member.

<u>Club Members Reflect on the Advantages of Club-Guided Projects</u>

The Freshman Orientation 2024, organized by the IC Design Club, was successfully held on October 19, 2024, at MV Hall, GJB. The event witnessed an enthusiastic participation of 120 first-year students.

Various fun and engaging activities were arranged, including crossword puzzles, quizzes, circuit design challenges, and other interactive games. The students actively participated and had an enjoyable experience.The event created a lively and welcoming atmosphere, allowing freshers to interact, learn, and compete in exciting challenges. Winners and participants received prizes as a token of appreciation for their efforts and enthusiasm.The orientation concluded at 1:30 PM, marking the beginning of an exciting journey for the new members of the club. The event was a great success, leaving students with wonderful memories and motivation for future club activities.

6)Treasure Hunt

The Treasure Hunt featured multiple levels, each filled with puzzles, clues, and obstacles that led teams closer to the hidden treasures. Participants had to decode tricky hints, solve mind-bending riddles, and work together efficiently to progress through the stages.Students thoroughly enjoyed the intensity and creativity of the challenges, making it not just a competition but also a fantastic opportunity for team bonding and collaboration.The event was more than just a game—it was a learning experience! Through teamwork and logical reasoning, students discovered the importance of communication, leadership, and coordination while racing against time

The event concluded with an exciting prize distribution, where the most skillful and strategic teams were crowned the ultimate treasure hunters! The enthusiasm and participation made the event a memorable experience for everyone involved.Special thanks to 5th semester student volunteers for arranging such a blissful event.

7)3D Printing workshop

The IC Design Club successfully organized the 3D Printing Workshop – 2024 on November 15th and 16th at the NRN Block IDEA Lab. This exclusive workshop, designed for first-year girls, provided an excellent opportunity to explore 3D modeling and printing while fostering creativity and technical skills.

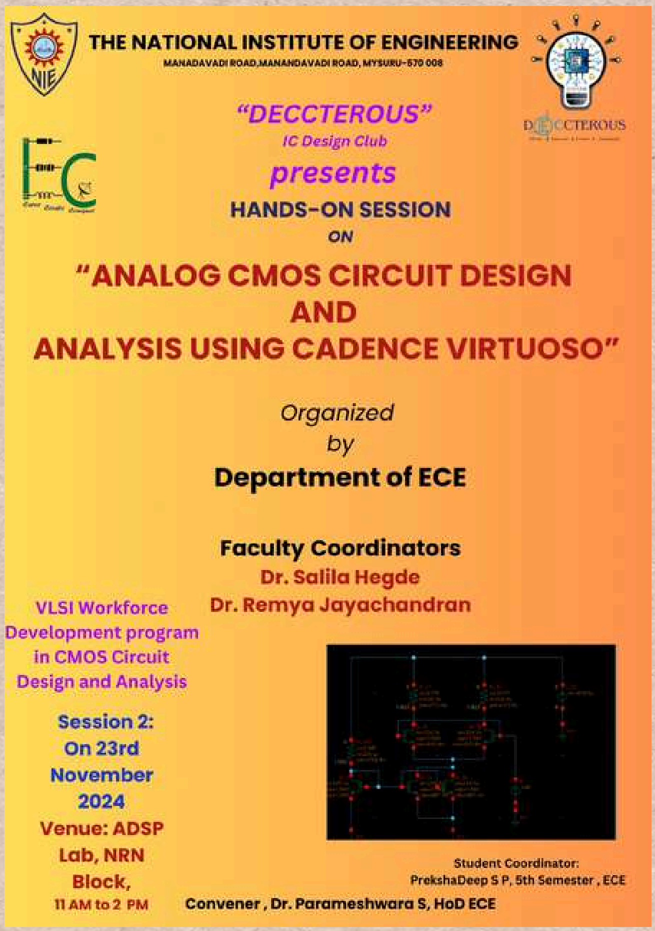
Hands-on Training: Participants learned the fundamentals of 3D design and printing techniques under expert guidance.Creative Learning Environment: The workshop encouraged innovation and problem-solving, making learning more engaging. Expert Guidance: The sessions were led by Dr. Anand A and Dr. Raghavendra T, esteemed faculty members from the Mechanical Engineering Department.

The workshop was meticulously planned and executed under the leadership of Dr. Remya Jayachandran and Dr. Salila Hegde, along with dedicated student coordinators from the NIE IW3DP Student Chapter and DECCTEROUS IC Design Club.The 3D Printing Workshop 2024 was a grand success, inspiring young minds to explore new-age technology and fostering a spirit of creativity and innovation

8)Hands-on Session on Analog CMOS Circuit

The IC Design Club, in collaboration with the Department of ECE, successfully conducted a hands-on session on "Analog CMOS Circuit Design and Analysis using Cadence Virtuoso" on November 23, 2024, at the ADSP Lab, NRN Block. This session was exclusively designed for 5th-semester students, aiming to enhance their knowledge and practical skills in CMOS circuit design and VLSI technology.The workshop, coordinated by Dr. Salila Hegde and Dr. Remya Jayachandran, provided students with in-depth training on Cadence Virtuoso, a widely used tool for circuit design, simulation, and analysis. Under the guidance of experts, participants learned about transistor-level design, performance optimization, and debugging techniques in analog circuits.Students actively engaged in designing and analyzing circuits, gaining valuable hands-on experience in VLSI design. The interactive nature of the session allowed students to ask questions, troubleshoot circuits, and explore advanced design methodologies.

The event was a great success, with enthusiastic participation from the students. It provided an excellent opportunity to bridge the gap between theoretical concepts and practical applications in the field of analog CMOS circuit design. The session concluded on a high note, leaving students motivated to explore further advancements in VLSI and semiconductor technology.



9) QUIZMOSIS

The IC Design Club, along with the Department of ECE, organized QUIZMOSIS, a four-day technical quiz competition from 13th to 16th November 2024. The event covered topics like aptitude, programming, analog electronics, digital logic, DSP, and communication systems.

Quiz Categories:

📌 Quiz 1: Aptitude & Physics

📌 Quiz 2: Mathematics & Programming

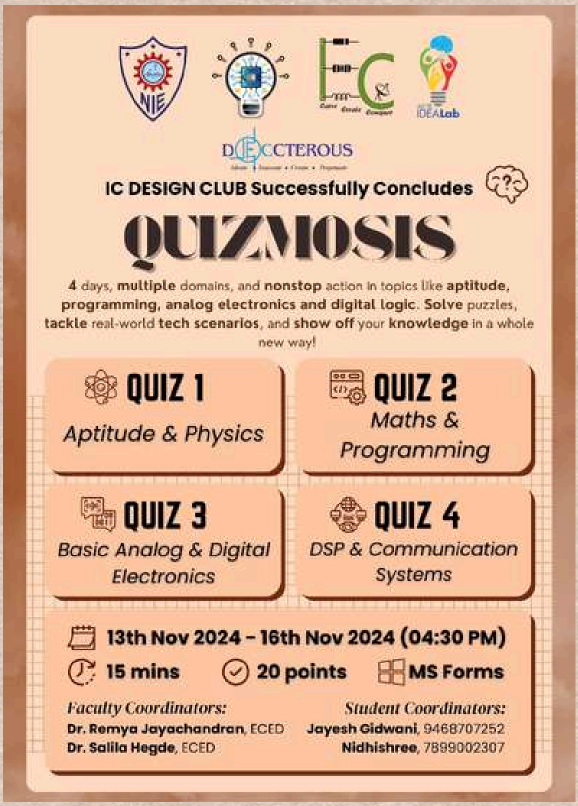
📌 Quiz 3: Basic Analog & Digital Electronics

📌 Quiz 4: DSP & Communication Systems

The quiz had 15-minute challenges with 20 points per quiz. The event was coordinated by Dr. Remya Jayachandran and Dr. Salila Hegde from the ECE Department. Jayesh Gidwani and Nidhishree, the student coordinators, helped in organizing the event.

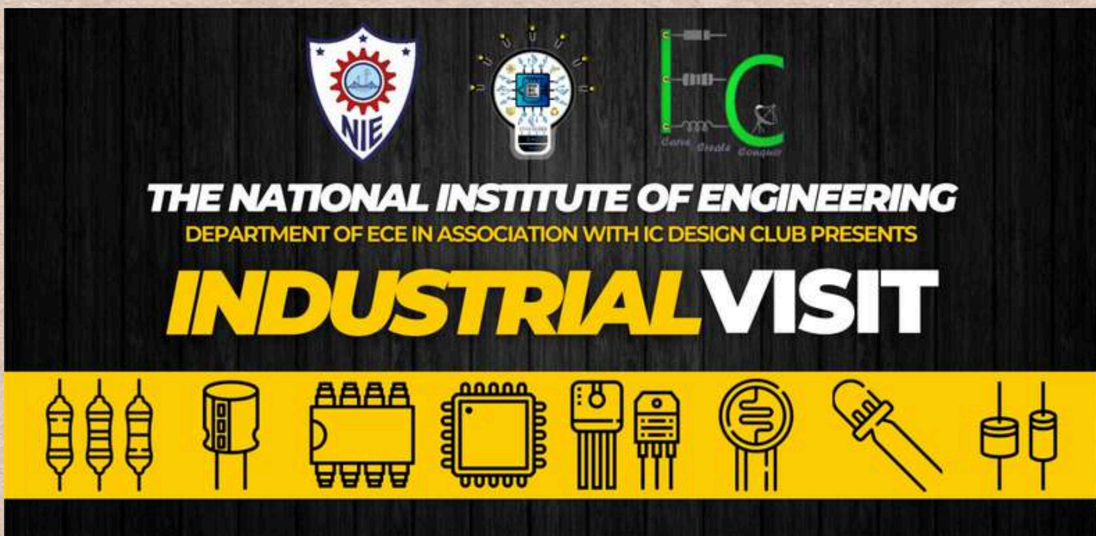
QUIZMOSIS Winners (November 2024)

Preethi (1st Sem ECE) ,Nesara N (1st Sem ECE),Thanushree S (1st Sem ECE),Suma Acharya (1st Sem EEE),Hemashree G N (3rd Sem ECE),Shashank G (3rd Sem EEE),Manjunath R (3rd Sem ECE), Mayuraditya J (5th Sem ECE), Sujala B R (5th Sem ECE)



10)Industrial visit

The IC Design Club, Department of ECE, NIE, organized a successful industry visit to Lahari Electronics, Mysore, on November 28, 2024, for third-semester ECE students. As part of the Analog Electronic Circuits course, this experiential learning initiative exposed 140 students to advanced testing and verification facilities like the EMI/EMC Chamber, Drop Test Chamber, and Environmental Test Chamber. Students interacted with industry professionals, gaining practical insights into electronics testing and quality assurance. The visit was made possible through the support of Bharati Sir, Lahari Electronics, and the committed efforts of student coordinators from the third semester and the IC Design Club team.We would also like to sincerely thank our dedicated IC Design Club student coordinators, Bheema Kashyapa, Jayesh G, Shreyansh, Visanth, and Satyadutt, for their hard work and commitment in making this event successful.The industry visit was a remarkable learning experience, helping students bridge the gap between theoretical knowledge and real-world applications.



Students at Lahari Electronics, Mysore

11)DIGITAL POSTER DESGIN

The Technical Digital Poster Design Competition was successfully conducted on November 30, 2024, at 10 AM in NRN Block by the Department of ECE. The competition allowed students to showcase their talent in website development by designing a website that displayed useful electronic components along with their part numbers used in PCB design. Participants creatively presented detailed technical information in an engaging and structured format. The jury evaluated the posters based on content accuracy, creativity, and technical relevance. A Q&A session was held where participants were asked technical questions related to the components, PCB part numbers, and website functionality. The jury panel consisted of Gurusatwik Bhattacharya, Project Associate at IISc Bengaluru, Amith Bharadwaj, R&D Engineer at Tejas Networks Ltd, and Divya S, Assistant Professor in the EEE Department.Faculty coordination was handled by Dr. Salila Hegde, Associate Professor in ECED, and Dr. Remya Jayachandran, Assistant Professor in ECED. The student coordinator for the event was Prekshadeep, Design Committee Lead. The event was a great success, with students displaying excellent technical knowledge and creativity, making it a valuable learning experience.

PRIZE WINNERS :

First prize with the cash prize of Rs. 2500:

Artistic Avengers.

2nd prize with the cash prize of Rs.1500:

Tech Transformers.

3rd prize with the cash prize of Rs. 1000:

Tech Art Designers.

Demostration of digital poster

Identification of Elecronic components

Group Photo of Participants and Mentors

Winners of the event

A token of appreciation for invaluable contributions

12)Analog Showdown

The Analog Showdown was successfully held on December 21, 2024, from 9 AM to 5 PM at NRN Block SES405, providing a platform for students to showcase their innovative ideas and circuit designs. The event aimed to ignite creativity and technical skills in electronics and communication engineering, with a keynote address on the Big Picture of Electronics and Communication Engineering by Sunil Shambhatnavar, CEO of Advanced Electronic Systems. Participants competed by designing and presenting analog circuits, demonstrating their expertise in circuit analysis, design, and problem-solving. The event was judged by a panel of esteemed evaluators, including Jayanth K (SoC Design Engineer, Intel India), Charan MS (Analog Layout Engineer, Texas Instruments), Prabhanjana (Software Developer Engineer, Siemens Bangalore), Sangeeta Shayana (Graduate Engineer Trainee, Schneider Electric), and Suhas Shastry (Associate Engineer, Morphing Machines). The competition was meticulously coordinated by Dr. Salila Hegde, Associate Professor, and Dr. Remya Jayachandran, Assistant Professor, ECE Department. The student coordinators, Disha L (Secretary) and Bheema Kashyapa (Projects & Industry Visit Committee Head), played a vital role in organizing the event.

PRIZE WINNERS :

📌Panel 1 (IDEA Lab): Smart Temperature Fan Control – Harsha B & Devi Prasad

📌Panel 2 (IoT Lab): Bind the Notes – Neha M N, Punyashree, Neethu

📌Panel 3 (AEC Lab): Motor Direction Control using H-Bridge – Pranav, Mahadev, Akash

📌Panel 4 (ADC Lab): 4:1 MUX using Discrete Components – Scholarson, Shreyansh, Satish, Sagar

📌Panel 5 (ASP Lab): Temperature Controlled DC Motor Fan – Shravani S, Sahana, Sadiya, Shwetha

📌 Special Award: Buck Converter – Surya, Hemanth, Nikhil, and Adarsh

📌Best Student Performance Award: Vishanth, Satyadatt, Vishwanath Sarapur, Pavan R, Harshitha B M, Sampreeth for their outstanding performance in The Analog Showdown 2024.

Token of appreciation to the resource person

Remya Mam addressing to the gathering

Judges evaluating projects

Judges giving valuable insights

ACHIEVEMENTS

On June 15, 2024, 23 members of the IW3DP NIE Student Chapter attended a hands-on 3D printing training at CIT Bangalore, in collaboration with IW3DP Bangalore, AMSI, EOS, and Additive Learning.The even is coordinated by Dr.Salila Hegde and Dr.Remya Jayachnadran, Mentors of IC designclub,ECE Department,NIE.

Group photo at CIT Banglore

Certificates of participants

Our students participated in the Student Research Symposium 2024 (SRS'24), organized by IIT Indore with IEEE. The theme was "Advancements in Emerging Technologies". Around 35 ECE students from NIE Mysore showcased their talents in various events. Harshitha B M won the Best Slide Design award, while Ananya D M, Dhanyashree D, Disha R S, and Anagha MK were Runners-up in Model Presentation.

Dr.Remya Jayachandran,Asst.Professor, ECE Department ,NIE was one of the session chair for the symposium.

Certificates of participants

The project students, Pundlik Anant, Raksha A R, Sangeetha Shayana, and Suhas Shastry under the guidance of Dr. Remya Jayachandran (DECCTEROUS – IC Design Club Faculty Coordinator) won the State Level "Best Project of the Year 2023-2024" Award by KSCST SPP 47 series with a cash prize of Rs, 6000/- and a funding of Rs. 5000/-. This is a significant achievement for these students, and it highlights their excellent work and contributions in the field of Electronics and Communication Engineering. This is a great milestone for the Department of Electronics and Communication Engineering at NIE Mysore

Award ceremony at KSCST SPP 47

The DECCTEROUS IC Design Club excelled at the National Level 24-Hour Circuit-A-Thon held at Dayananda Sagar College of Engineering, Bangalore. Competing in the Analog Design category, our teams secured:

Third Prize: S Sudeep Kumar, Varshini H R, Sanjay H, and Prekshadeep for their Memristor Emulator Circuit Design for Neuromorphic Computing.

Special Jury Appreciation: Harshitha B H, Satyadatt M S, and Neha M N for their Low Dropout Voltage Regulator project.

Guided by Dr. Remya Jayachandran, their hard work and innovation earned well-deserved recognition.

Students receiving award at DSCE Bangalore

Students from NIE secured Second Prize in the "STUDENT INNOVATIONS LIVE: TECH EXHIBITS" category at the WITINFLUENZE Conference 2024, organized by WEquity for Women and Technology at Radisson Blu, Bangalore on September 26, 2024.

Nidhishree, a third-semester ECE student, won Second Place and a cash prize of ₹7,500 for her project, "Automated Plastic Segregation System Based on Grade Classification Using Arduino," developed under the guidance of Dr. Remya Jayachandran. The project was also shortlisted for FAER 2023-'24, mentored by Dr. Chanakya Hosall, Chief Research Scientist at IISc.

A total of 46 students and 4 faculty members participated in the conference, an initiative by the IC Design Club to promote Women Empowerment, coordinated by Dr. Salila Hegde, Dr. Remya Jayachandran, and Dr. Rajeshwari S.

Our students at WITINFLUENZE Conference 2024

We extend our heartfelt gratitude to Dr. Remya Jayachandran for her exceptional leadership as the lead of the Communication and Publicity team in Smart India Hackathon (SIH) 2024. Her guidance and unwavering support were instrumental in ensuring the success of our volunteers—Anagha MK, Harshitha BM, Neha H S, and Ananya D M—in carrying out their responsibilities efficiently.

SIH 2024 MYSORE

The One-day workshop on Environment and Green Energy: PARYAVARAN-2024 was organized by the INAE Bangalore Chapter in collaboration with Gokula Education Foundation. The One-day workshop on Environment and Green Energy: PARYAVARAN-2024 was organized by the INAE Bangalore Chapter in collaboration with Gokula Education Foundation. This event focuses on raising awareness and promoting innovative solutions related to environmental conservation and sustainable energy. Key topics include: - Renewable energy technologies and their impact on reducing carbon footprints. - Discussions on environmental protection and sustainable development strategies. Sujala B R, Thanushree M S, Reefa Tamaniyat participated in this competition and workshop

Guided Project

1. Design of MIF cell using CMOS memristor emulator circuit - Sudeep Kumar, Varshini H R, Prekshadeep, Sanjay H
Guide: Dr. Remya Jayachandran, External Guide: Sangeetha A Shayana
Phase 1 completed, Publication : 1 (accepted), Third Prize for Circuit-A -Thon Competition in Analog Domain

2. Jaundice Detector- Yash Nema, Saloni Kumari, Scholarson
Guide: Dr. Remya Jayachandran, External Guide: Dr. Chanakya H , IISc
Phase 1 completed, Funding : FAER 2024-25 (ongoing), First Prize for IDT Project

3. Design of Phase detector & Charge pump for PLL in 5G applications- Disha L, Chandrashekar, Ankitha, Ksheeraj
Guide: Dr. Remya Jayachandran, External Guide: Charan M S
Phase 1 ongoing

4. Automatic Plastic waste segregation system for MRCs - Nidhishree
Guide: Dr. Remya Jayachandran, External Guide: Dr. Rupesh S, BRIDC, Dr. Chanakya H
Completed, Publication: 1 (presented), Best Paper award, Second prize with Cash Prize in WEQUITY 2024, Funding FAER 2023-’24

Research Publications 2024-2025

<u>Research work of ICDC Faculty Coordinators with students of IC Design Club</u>

1. Vas, Maelona Maxim, A. Shriranjani, B. C. Nishchith, Rajalekshmi Kishore, and Salila Hegde. "Design and Development of a Functional CANSAT Model for Atmospheric Data Collection." In 2024 IEEE Space, Aerospace and Defence Conference (SPACE), pp. 1204-1207. IEEE, 2024.

2.Remya Jayachandran, Edet Bijoy Kumaradhas, Arun S, "Design of second order low pass filter using inverter-based CMOS operational transconductance amplifier (OTA) through gm/ID Methodology", 2024 IEEE International Conference of Electron Devices Society Kolkata Chapter (EDKCON), Kolkata, India, 2024.

3.Pundlik Anant Nayak, Raksha A R, Sangeetha Shayana, Suhas Shastry, Remya Jayachandran, "A Compact and Low power memristor based 8-bit digital to Analog converter for hearing Aid devices", 10th International Conference, IEEE CONECCT July 2024

4.Remya Jayachandran, Rupesh S, Shaen Kalathil, "Intelligent Solid Waste Sorting System using IoT and Image processsing ", IEEE 2024 4th Asian Conference on Innovation in Technology (ASIANCON), Pune , Maharashtra ,India

5.Jayachandran, Remya, Bhargava Dantapur, Aneeta S. Antony, and Rohini Nagapadma. "Transforming Healthcare Through Smart Health Systems: Harnessing Technology for Enhanced Patient Care." In Transforming Healthcare Sector Through Artificial Intelligence and Environmental Sustainability, pp. 107-127. Singapore: Springer Nature Singapore, 2025.

6.Rupesh, S., S. L. Aravind, B. R. Kavya, and Remya Jayachandran. "Recent advances and new concepts in methane storage and transportation." Advances and Technology Development in Greenhouse Gases: Emission, Capture and Conversion (2024): 323-354.

7.Remya Jayachandran, Nidhishree, Anush R Halappanavar, Laxman K Budni, Abhishesk C M, Charan S Hegde, and Rupesh S, "SUSTAINABLE AUTOMATED PLASTIC WASTE SEGREGATION SYSTEM USING IOT AND IMAGE PROCESSING FOR MATERIAL RECOVERY CENTERS", Second International Conference on Sustainable Technologies (ICST) 2025. (presented) - Best Paper Award 2025

8.Remya Jayachandran, Sudeep Kumar, Sanjay H, Varshini H R, Prekshadeep S P, Sangeetha A Shayana, "CMOS-Based Memristor Emulator Circuit for Audio Signal Processing in Hearing Aid Applications", In 6th IEEE International Conference on Devices for Integrated Circuits (DevIC 2025), April 2025 held at Kalyani Government Engineering College, Kolkata. (Accepted).

Article section

Cybersecurity and IoT Sensors Integrated Research

The Internet of Things (IoT) has revolutionized industries by enabling smart, interconnected devices that collect, transmit, and process data in real time. IoT sensor networks are accessed across critical applications such as healthcare, smart cities, industrial automation, and autonomous vehicles. These sensors facilitate intelligent decision-making and automation, improving efficiency and user experience. However, the rapid expansion of IoT ecosystems has introduced new cybersecurity vulnerabilities, making them targets for cyberattacks. Unlike traditional IT systems, IoT devices often operate in resource-constrained environments with limited processing power, storage, and security mechanisms. Many IoT sensors lack robust authentication protocols, making them susceptible to threats such as unauthorized access, data breaches, denial-of-service (DoS) attacks, and false data injection. Additionally, IoT networks rely on wireless communication, exposing them to man-in the-middle (MITM) attacks, passive attacks, and signal jamming.

To address these challenges, advanced cybersecurity strategies are required to secure IoT sensor networks. Like the groundbreaking research – Anomaly Detection Approach of using AI-powered systems to detect cyber threats in IoT sensors. Since traditional security methods are not suitable for IoT due to limited device resources, this system uses machine learning to spot unusual sensor activity in real time. The method is lightweight and efficient, ensuring quick threat detection and response without slowing down the network. By running detection at the edge (closer to the sensors), this approach improves security while reducing delays. Other research also integrates Quantum and Cryptography technologies to solve the cyberthreats.

As IoT adoption continues to grow, strengthening cybersecurity frameworks will be essential to protecting smart homes, healthcare systems, industrial automation, and critical infrastructures. Future research should focus on developing scalable, energy efficient, and adaptive security solutions to keep pace with evolving cyber threats, ensuring a safe and resilient IoT ecosystem.

Suma Acharya

USN: 4NI24EE108

BRANCH: Electrical & Electronics

UNDERSTANDING THE ELECTROMIGRATION EFFECT IN IC DESIGN

It is crucial for students to understand concepts beyond the syllabus. Here, I'm introducing an interesting topic which will be helpful for those who target to enter into the VLSI domain.

In practical design implementation, various non-ideal effects come into play. This article introduces the impact of electromigration in integrated circuit (IC) design. As you may know, the flow of charge carriers generates current in a semiconductor device. As device dimensions shrink, the number of interconnect layers increases. Electrons must reach the outside world without delay. However, due to the high velocity of electrons moving through these thin metal layers, metal corrosion can occur, especially at sharp corners. This can lead to signal detachment within the circuit.

In this article, we will briefly discuss interconnects and the effects of electromigration.

In IC design, interconnects are essential for connecting transistors, logic gates, and functional blocks. As device scaling advances into the nanometer regime, interconnects significantly impact performance, power consumption, and reliability. Efficient interconnect design is crucial for achieving high-speed and low-power operation in modern ICs. The interconnects and the vias are depicted in Fig. 1.

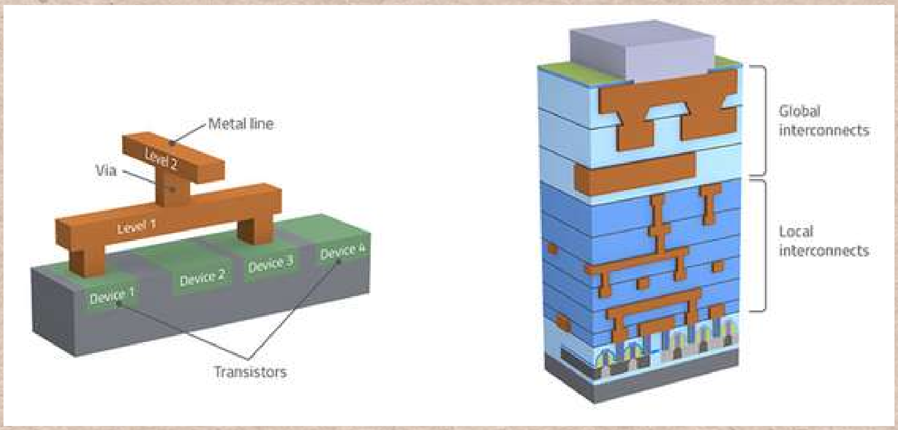


Figure 1

Key Challenges in Interconnect Design

1)RC Delay (Resistance-Capacitance Delay)

As feature sizes shrink, interconnect resistance and capacitance increase, degrading signal speed.

Use low-k dielectrics and wider wires for critical paths.

2)Electromigration

High current density causes metal atoms to migrate, leading to failure.

Use redundant vias, wider interconnects, and Cu-based materials.

3)Crosstalk & Signal Integrity

Close interconnect spacing increases capacitance, causing signal interference.Introduce shielding layers and optimize wire spacing.

As semiconductor technology scales down to nanometer dimensions, electromigration (EM) has become a critical reliability concern in IC design. This phenomenon, driven by high current densities, can degrade metal interconnects over time, leading to circuit failure. Understanding and mitigating electromigration is essential for ensuring long-term reliability in modern ICs. Electromigration is the gradual movement of metal atoms in a conductor due to momentum transfer from flowing electrons. This effect occurs in high-current-density regions, particularly in metal interconnects made of aluminum (Al) or copper (Cu). Over time, electromigration can cause two major failure modes:

1. Void Formation: Leads to increased resistance and potential open circuits.
2. Hillocks and Extrusions: Can result in short circuits between adjacent interconnects.

Voids and Hillock formation is shown in Fig. 2. The TEM image of the interconnects is presented in Fig. 3. and Fig. 4

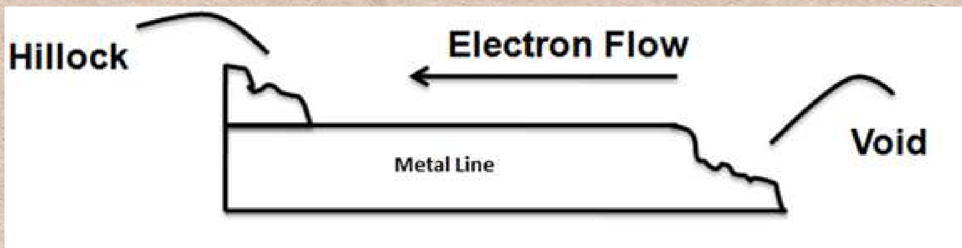


Figure 2

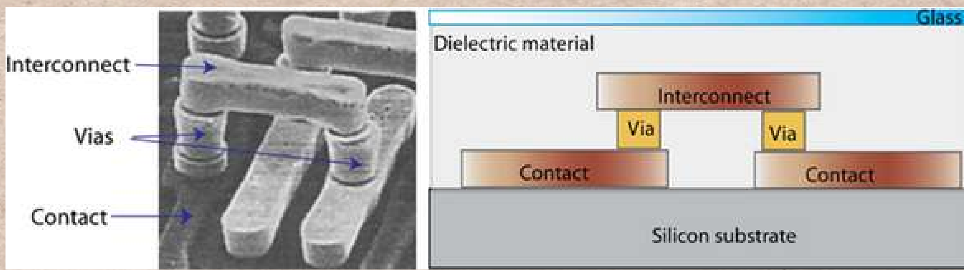


Figure 3

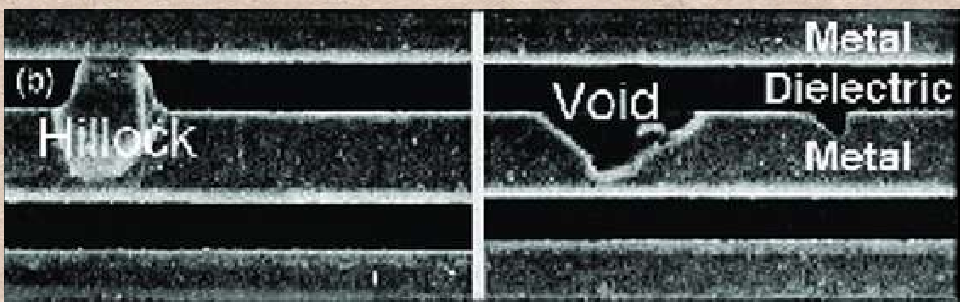


Figure 4

Electromigration is influenced by several factors:

· Current Density: Higher current densities accelerate atomic displacement.

· Temperature: Elevated temperatures increase atom mobility, exacerbating EM.

· Material Properties: Copper has better EM resistance than aluminum due to its stronger atomic bonding.

· Interconnect Geometry: Narrower wires experience higher current density, making them more susceptible to EM.

To enhance IC reliability, designers employ various techniques to mitigate electromigration:

1. Wider Interconnects: Reducing current density per unit area.
2. Redundant Vias: Providing alternative current paths to prevent open circuits.
3. Barrier Layers: Using materials like TiN, TaN, or Co to block atom diffusion.
4. Low-Resistance Metals: Replacing aluminum with copper interconnects, which have better EM resistance.
5. Current Density Optimization: Ensuring circuit design keeps current levels within safe limits.

Electromigration poses a significant challenge in Advanced IC design, particularly in nanometer-scale technologies. By understanding its mechanisms and implementing proper design techniques, engineers can mitigate its effects and enhance the reliability of modern integrated circuits.

References:

1. Plummer, James D. Silicon VLSI technology: fundamentals, practice and modeling. Pearson Education India, 2009.
2. Lienig, J. (2013, March). Electromigration and its impact on physical design in future technologies. In Proceedings of the 2013 ACM International symposium on Physical Design (pp. 33-40).
3. Thompson, C. V., & Lloyd, J. R. (1993). Electromigration and IC interconnects. Mrs Bulletin, 18(12), 19-25.
4. Lienig, Jens, Susann Rothe, and Matthias Thiele. "Mitigating electromigration in physical design." In Fundamentals of electromigration-aware integrated circuit design, pp. 107-155. Cham: Springer Nature Switzerland, 2025.
5. Wang, Albert. Practical ESD Protection Design. John Wiley & Sons, 2022.

Article By:
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Everything about C Pointers

Introduction

Pointers are one of the most powerful and essential features in the C programming language. They provide a way to directly manipulate memory, allowing for efficient and dynamic memory management. Pointers often lead to more compact and efficient code that can be obtained in other ways. Understanding pointers is crucial for working with arrays, functions, and dynamic memory allocation.

What is a Pointer?

Let us understand how memory is organized with the following diagram. Typically, a machine has an array of consecutively addressed memory cells that can be individually manipulated or in contiguous groups. Normally a memory location can hold 1 byte of data. A pointer is a variable that stores the memory address of another variable. The contents of the memory location can be accesses by dereferencing the pointer which is known as indirect addressing mode.

Syntax of Pointers

To declare a pointer in C, use the * operator as int *ptr; Here, ptr is a pointer variable to an integer.

Initializing a Pointer

A pointer must be assigned the address of a variable before it can be used. This is done using the & (address-of) operator. int a = 10; int *ptr = &a; Now, ptr stores the address of a. and ptr is said to “point to” a. This is known as referencing the pointer using operator &. The & operator applies only to objects in memory : variables and array elements. It cannot be applied to expressions, constants or register variables.

Dereferencing a Pointer

To access the value stored at the address a pointer is pointing to, use the * operator: printf("Value of a: %d", *ptr); /* prints 10*/ This prints the value of a using its pointer. The unary operator * is the indirection or dereferencing operator. Following is another illustration for pointer referencing and dereferencing operations. int x=1,y=2,z[10]; int *ip; ip =&x; y=*ip; *ip =7; ip =&z[0]; the declaration of the pointer ip , int *ip is intended as a mnemonic; it says expression *ip is an int. Similarly the expression,
double *dp; says that in an expression *dp has value double. If ip points to integer x, then *ip can occur in any context where x could, so *ip =*ip +10; increments *ip by 10 i.e. x by 10.

Pointer Arithmetic Since pointers store memory addresses, arithmetic operations can be performed on them: ptr++; // Moves to the next memory location ptr--; // Moves to the previous memory location

```
EX: Int x=5,y=7; int *xp =&x; int *yp=&y;
    *xp = *xp +1; /* *xp (i.e. x) increments by 1 so x=6 */
     xp = xp +1; /* xp increments by 1 , xp points to next address */
     yp = yp-1; /* yp address decrements
     by 1 */
     xp ++;
```

yp--;

++*xp;

(*yp)++;

*ip++;

If ip and iq are pointers to integers then iq =ip is valid;iq points to the variable where ip is poining

Pointers and Arrays

An array name in C acts as a pointer to its first element.Therefore, arrays and pointers are closely related.

int arr[] = {1, 2, 3, 4, 5};

Array name arr is a pointer itself and always point to the first element of array hence,

int *ptr = arr; // Points to the first element of arr

ptr =&arr[0] ; //points to the first element of array

You can access array elements using pointers:

printf("%d", *(ptr + 1)); // Prints second element of arr

ptr = ptr+1; // now ptr points to second element of array

arr = arr+1 ; // Not valid array pointer can not be incremented or decremented, Since it is a constant pointer.

ptr is variable pointer and arr is constant pointer. ptr+i refers to address of arr[i]. Reference to arr[i] can also be written as *(arr+i). i.e., & arr[i] and (arr+i) are equal. Hence ptr[i] is same as *(ptr+i).

Constant pointer is the one whose value (memory address it holds) cannot be changed after initialization, meaning it always points to the same memory location. It is declared as data_type *const pointer_name;

EX: int a,b ; int *const ptr = &a;

An array pointer is an example of constant pointer.

Pointers to constant

This means the pointer can point to different memory locations , but the value it points cannot be changed through pointer.

It is declared as,

const int a=10; int b;

const int *ptr1 =&a ; * valid as a is constant*\

ptr1 = &b; * not valid as b can have different values *\

We can combine both concepts as

const int a =20; int b;

const int * const ptr= &a ; * ptr is a consant pointer which points to a constant value*\

Declarations like

const int * const ptr =&b is not valid as b is a variable not a constant.

Using const helps prevent accidental modification of data or memory locations improving code reliability and safety.

Pointers and Functions

Pointers can be passed to functions to achieve call-by-reference.

```
int main() {
void modify(int *x) {
 *x = 20;)}
```

ptr always points to address of variable a. The statements like ptr = &b; or ptr++; or ptr =ptr+2 etc are not valid.

```
int a = 10;
modify(&a);
 printf("%d", a); // Output: 20 return 0; }
```

Here, a is modified inside the function using a pointer.

Dynamic Memory Allocation

C provides functions such as malloc, calloc, and free to dynamically allocate and deallocate memory.

```
 int *ptr = (int *)malloc(sizeof(int) * 5);
 if(ptr == NULL) {
 printf("Memory allocation failed");
}
 else {
// Use allocated memory
free(ptr); // Deallocate memory
}
```

Null and Void Pointers

Null Pointer: A pointer that does not point to any valid address.

```
int *ptr = NULL;
```

Void Pointer: A generic pointer that can store any data type's address.

```
void *ptr; int x = 10;
 ptr = &x;
```

Any pointer to an object can be converted to type void * without loss of information If the result is converted back to the original pointer type the original pointer is recovered.

Conclusion

Pointers are a fundamental part of C programming, providing direct access to memory and improving efficiency. Mastering pointers allows for effective manipulation of data structures like arrays, linked lists, and trees, as well as optimizing memory usage in complex programs.

Article by:
Dr.Salila Hegde,
Associate Professor,
ECE,NIE

Acknowledgment

We extend our heartfelt gratitude to all the participants, evaluators, and attendees who contributed to the success of these remarkable events. Your enthusiasm, dedication, and innovative spirit made each competition a platform for creativity and learning.

A special thank you to our keynote speakers, mentors, and industry experts for sharing their valuable insights and guiding aspiring engineers toward excellence. Your encouragement and expertise have played a crucial role in shaping the future of technology.

We sincerely appreciate the support of The National Institute of Engineering, Mysuru, along with the faculty coordinators, including Dr. Salila Hegde, Dr. Remya Jayachandran, and all the mentors, for their unwavering encouragement and mentorship. Your guidance has been instrumental in nurturing talent and fostering a culture of innovation.

To the organizing team, student coordinators, and volunteers—your hard work, dedication, and seamless execution of these events have ensured their grand success. Your efforts are truly commendable!

Finally, a huge congratulations to all the winners and participants for their outstanding performances. Your passion and commitment to engineering excellence continue to inspire us all.

We look forward to many more such opportunities to ignite ideas and create innovations together!

Best Regards,

DECCTEROUS -IC Design Club,

The National Institute of Engineering, Mysuru

Acknowledgment from Faculty coordinators

As a faculty mentor of the IC Design Club, I would like to express my sincere gratitude to all those who have contributed to the growth and success of this initiative. First and foremost, I extend my heartfelt thanks to the Management, Principal Dr. Rohini Nagapadma and Dr. Parameshwara S, Head of department of ECE for their continuous encouragement and support in fostering innovation in Integrated Circuit (IC) Design. Their vision and commitment have enabled us to create a platform for students to explore and excel in this domain.

I would also like to acknowledge the hard work and dedication of our student members. Your enthusiasm, curiosity, and willingness to learn have been the driving force behind the club's success. Your participation in workshops, projects, and technical discussions reflects your passion for IC design and strengthens our mission.

A special note of appreciation to my fellow faculty mentor Dr.Remya Jayachandran who is instrumental in taking initiative and setup club with hard work, sincere efforts and dedication. I also acknowledge the fellow faculty members and colleagues who have provided their valuable insights, technical expertise, and support in guiding the students. Your contributions have been instrumental in shaping the club's initiatives and activities.

I am also grateful to our Alumni, Industry professionals, guest speakers, and collaborators who have shared their knowledge, real-world experiences, and mentorship. Your support has helped bridge the gap between academia and industry, providing our students with practical exposure and career opportunities. Lastly, I appreciate the efforts of all coordinators, volunteers, and organizing members who have worked behind the scenes to ensure the smooth execution of various events and activities.

Your dedication and teamwork have played a crucial role in the club's success. I look forward to seeing the IC Design Club continue to grow, innovate, and inspire future engineers in the field of VLSI and semiconductor design. Thank you all for your support and commitment!

With regards

Dr.Salila Hegde,

Associate professor,

Faculty mentor, DECCTEROUS - IC design club,

ECE, NIE

Acknowledgment from Faculty coordinators

DECCTEROUS IC DESIGN CLUB: PIONEERING THE FUTURE OF MULTIDISCIPLINARY ENGINEERING

"The only way to achieve the impossible is to believe it is possible." – Charles Kingsleigh

The DECCTEROUS IC Design Club is a student-driven, faculty-mentored initiative dedicated to advancing knowledge in semiconductor technology, IC design methodologies, embedded systems, programming, AI-ML, and more. Officially inaugurated on July 6, 2024, the foundation for this club was laid as early as 2020-21, fueled by the passion and perseverance of our student community. Serving as a dynamic platform, DECCTEROUS empowers students with technical expertise, hands-on project experience, and insights into industry advancements while strengthening fundamental concepts. Over the past nine months, the club has actively conducted workshops, training sessions, guest lectures, design challenges, technical competitions, women empowerment programs, industrial visits, collaborative projects, RISC-V programs, and more. These activities ensure students remain engaged with cutting-edge technologies and gain invaluable practical exposure.

The DECCTEROUS IC Design Club has played a vital role in shaping future engineers, equipping them with skills in circuit design, system-level integration, debugging, PCB design, and embedded programming. Many members have secured prestigious internships, excelled in hackathons, presented research papers, undertaken industry-sponsored projects, and emerged as winners in technical competitions—all under the guidance of dedicated faculty mentors. I extend my heartfelt congratulations to the Office Bearers of DECCTEROUS IC Design Club for AY 2024-25 and all student members of ICDC for their exceptional achievements, active participation, and commitment to innovation. Your dedication has been instrumental in the club's success, and I hope you continue to inspire, lead, and push the boundaries of semiconductor technology and engineering. As the Faculty Coordinator, I take immense pride in witnessing the growth and enthusiasm of our students in multidisciplinary engineering. DECCTEROUS is more than a technical forum—it is a thriving community of innovators, collaborators, and passionate learners. Our mission is to equip students with cutting-edge skills and industry insights, preparing them to become future leaders in engineering. Whether you're a beginner or an advanced learner, the club offers a wealth of opportunities to explore, create, and innovate. "Success is no accident. It is hard work, perseverance, learning, studying, sacrifice, and most of all, love of what you are doing." – Pelé

Be a part of a community that is shaping the future.

"The woods are lovely, dark and deep, But I have promises to keep, And miles to go before I sleep, And miles to go before I sleep." – Robert Frost .

Try hard to achieve your goals.

Best wishes to all my dear students!

Dr. Remya Jayachandran, DECCTEROUS-IC Design Club, Faculty Coordinator

Acknowledgment from Faculty coordinators

“Engineering or technology is all about using the power of science to make life better for people, to reduce cost, to improve comfort, to improve productivity, etc.” – N. R.Narayana Murthy

First and foremost, I extend my heartfelt thanks to the Management, Principal Dr. Rohini Nagapadma and Dr. Parameshwara S, Head of department of ECE for their continuous encouragement and support in fostering innovation in Integrated Circuit (IC) Design. Their vision and commitment have enabled us to create a platform for students to explore and excel in this domain.

“The engineer’s first problem in any design situation is to discover what the problem really is.”

As a faculty mentor of the IC Design Club, I would like to appreciate the active involvement of our students with their sincere hard work and dedication in all the activities and in identifying real world problems to give their innovative solutions through their project implementation. Your enthusiasm, curiosity, and willingness to learn have been the driving force behind the club’s success. Your participation in workshops, projects, and technical discussions reflects your passion for IC design and strengthens our mission.

A special note of appreciation to my Senior faculty mentors Dr. Salila Hegde and Dr. Remya Jayachandran who are back bone in taking initiative and setup club with sincere efforts and dedication. I also acknowledge the fellow faculty members and colleagues who have provided their valuable insights, technical expertise, and support in guiding the students. Your contributions have been instrumental in shaping the club’s initiatives and activities.

I am also grateful to our Alumni, Industry professionals and Guest speakers who have shared their knowledge, real-world experiences, and mentorship. Your support has helped bridge the gap between academia and industry, providing our students with practical exposure and career opportunities in VLSI, Semiconductor industry, Embedded system programming and many more.

Lastly, I appreciate the efforts of all coordinators, volunteers, and organizing members who have worked behind the scenes to ensure the smooth execution of various events and activities.

Thank you all for your support.

I look forward to seeing the IC Design Club continue to grow, innovate, and inspire future engineers in the field of VLSI and semiconductor design.

“Success is not final; failure is not fatal: It is the courage to continue that counts.” -Winston Churchill

Best wishes to our innovative younger minds who are going to build their creative future.

Regards,

Pushpalatha H P

Assistant Professor

Dept. of ECE

Acknowledgment from Faculty coordinators

As a faculty mentor of the IC Design Club, I would like to express my sincere gratitude to all those who have contributed to the growth and success of this initiative.

First and foremost, I extend my heartfelt thanks to the Management, Principal Dr. Rohini Nagapadma and Dr. Parameshwara S, Head of department of ECE for their continuous encouragement and support in fostering innovation in Integrated Circuit (IC) Design. Their vision and commitment have enabled us to create a platform for students to explore and excel in this domain. I would also like to acknowledge the hard work and dedication of our student members. Your enthusiasm, curiosity, and willingness to learn have been the driving force behind the club's success.

Your participation in workshops, projects, and technical discussions reflects your passion for IC design and strengthens our mission. A special note of appreciation to my fellow faculty mentor Dr.Salila Hegde who is instrumental in taking initiative and setup club with hard work, sincere efforts and dedication. I also acknowledge the fellow faculty members and colleagues who have provided their valuable insights, technical expertise, and support in guiding the students. Your contributions have been instrumental in shaping the club's initiatives and activities.

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Lastly, I appreciate the efforts of all coordinators, volunteers, and organizing members who have worked behind the scenes to ensure the smooth execution of various events and activities. Your dedication and teamwork have played a crucial role in the club's success.

I look forward to seeing the IC Design Club continue to grow, innovate, and inspire future engineers in the field of VLSI and semiconductor design.

Thank you all for your support and commitment!

Regards,
Pawan Bharadwaj
Assistant Professor
Dept of ECE , NIE

Acknowledgment from President

Dear Members,

I am honored to extend my heartfelt gratitude to everyone who contributed to the successful completion of our 2024 Club Newsletter. This publication stands as a testament to the dedication, creativity, and teamwork demonstrated by our members.

We would like to express our sincere appreciation to the editorial team for their hard work in curating and designing the content, as well as to all the contributors who shared their valuable insights, articles, and creative pieces. Your efforts have made this newsletter an engaging and informative platform that reflects the spirit and vision of our club.

A special thank you to our faculty advisors, sponsors, and supporters for their unwavering guidance and encouragement throughout the process. Your contributions have played a pivotal role in the success of this endeavour.

Let this newsletter serve as a source of inspiration and motivation for future initiatives. We are confident that the same passion and commitment will drive our club to greater achievements in the coming years. Thank you once again for your invaluable support and participation.

Warm regards,
Sujala B R
President, IC Design Club – Deccterous,
Department of ECE,NIE Mysore

Acknowledgment from Vice President

Dear Members,

I am delighted to extend my heartfelt gratitude to everyone who has contributed to the successful completion of our 2024 Club Newsletter. A special thanks to the news letter creation team whose dedication has helped us in creating this newsletter. I would also extend my gratitude to each and every person involved in making this newsletter. I also congratulate all those who have submitted their articles. Your contributions have made this newsletter an informative source. I would also like to express my gratitude towards our faculty advisors Remya Mam and Salila Mam and supporters for their continuous guidance and encouragement. Your unwavering support has played a vital role in shaping this club. Once again, thank you all for your invaluable efforts and support.

Warm regards,

Sanjay S Hanchinal

Vice President, IC Design Club - Deccterous

Department of ECE, NIE Mysore

Acknowledgment from Secretary

Dear Members,

I am delighted to extend my gratitude to everyone who played a part in the successful launch of our 2024 Club Newsletter. As your secretary, I have had the privilege of witnessing the dedication, creativity, and teamwork that each of you brought to this endeavor.

Our editorial team deserves special recognition for their tireless efforts in curating and designing the content. To all contributors—thank you for sharing your invaluable insights, technical articles, and creative pieces. It is your contributions that truly reflect the spirit and excellence of our club. A heartfelt thank you goes out to our faculty advisors, sponsors, and mentors, whose guidance and support have been integral to the success of this publication.

Your encouragement continues to inspire us. I hope this newsletter serves as a testament to the achievements of the past year and motivates us for future initiatives. Let's continue to push the boundaries of innovation, learning, and collaboration in the field of integrated circuits and design. Thank you all once again for your unwavering support and contributions.

Warm regards,

Disha L

Secretary, IC Design Club – Decctrous,

Department of ECE, NIE Mysore

Acknowledgment from Treasurer

Dear Members,

I am honoured to express my sincere appreciation to everyone who contributed to the successful release of our 2024 Club Newsletter. As your Treasurer, I have witnessed the hard work, dedication, and collaboration that made this publication possible.

A special thanks to our editorial team for their relentless efforts in curating engaging content. To all contributors—your technical insights and creative articles have truly enriched this newsletter, making it a reflection of our club's excellence.

I extend my gratitude to our faculty advisors, sponsors, and mentors, whose invaluable support has played a crucial role in this endeavour. Your guidance continues to inspire us to strive for greater heights. This newsletter is a testament to our achievements and a stepping stone for future innovations. Let's continue fostering learning, creativity, and excellence in the field of integrated circuits and design. Thank you all once again for your unwavering support and dedication.

Warm regards,

SHESHA KRISHNA S
Treasurer, IC Design Club – Decctrous,
Department of ECE, NIE Mysore

Acknowledgment from other committees

Dear Members,

As the Event Organizing Committee of the IC Design Club – Deccterous, we take immense pride in acknowledging the successful execution of our 2024 Club Newsletter event. It was a wonderful journey of meticulous planning and seamless coordination to bring this initiative to life. Our goal was to ensure a well-structured, engaging, and smooth event that celebrated the hard work of all involved. Through dedicated teamwork and careful organization, we aimed to create an atmosphere that reflected the enthusiasm and commitment of our club. We extend our sincere gratitude to the Editorial Committee for their vision and well curated content, which laid a strong foundation for this event.

A heartfelt thanks to the Design Committee for their outstanding creative efforts in making the newsletter visually appealing. Additionally, we deeply appreciate the unwavering support from our faculty advisors, whose valuable guidance played a crucial role in our success. This achievement stands as a testament to the collaborative spirit of our club. We hope this milestone serves as inspiration for future members to take up new challenges with the same energy and dedication. Thank you for your trust in us and for the opportunity to contribute to this incredible journey.

Warm regards,

Thanushree M S & Shrilakshmi

Event Organizing Committee IC Design Club – Deccterous Department of ECE,

NIE Mysore

Dear Members,

As the Website Committee of the "lC Design Club – Deccterous", we take immense pride in acknowledging the successful execution of our "2024 Club Website Revamp". This journey has been a blend of meticulous planning, technical expertise, and seamless collaboration to bring dynamic and engaging platform to life. Our goal was to ensure a well-structured, visually appealing, and user-friendly website that reflects the essence of our club. Through dedicated teamwork and careful organization, we aimed to create a digital space that enhances accessibility, showcases our initiatives, and fosters engagement among members.

We extend our sincere gratitude to the "Content Team" for their valuable insights and well crafted write-ups, which added depth to our website. A special thanks to the "Design Team" for their creative efforts in ensuring an aesthetically pleasing interface.

Additionally, we deeply appreciate the unwavering support from our "faculty advisors", whose guidance played a crucial role in shaping this project. This achievement is a testament to the "collaborative spirit" of our club. We hope this milestone inspires future members to embrace new challenges with the same enthusiasm and dedication.

Thank you for your trust in us and for the opportunity to contribute to this incredible initiative.

Warm regards,

Parushuram M & Sagar Kumar Singh

Website Committee IC Design Club – Deccterous

Acknowledgment from other committees

Dear Members,

As the Stage Committee of the IC Design Club – Deccterous, we take immense pride in our commitment to the smooth conduction and structured execution of events. The 2024 Club Newsletter Event was a testament to this dedication, and we are delighted to have played a key role in its seamless coordination. Our focus was to ensure that every aspect of the event unfolded efficiently, creating an engaging and well-organized experience for all. Through meticulous planning and teamwork, we strived to uphold the club's standards of excellence and professionalism. We extend our sincere gratitude to the Editorial Committee for their insightful and well-curated content, which formed the heart of this event. A special thanks to the Design Committee for their exceptional creativity in making the newsletter visually captivating. We also deeply appreciate the invaluable support and guidance of our faculty advisors, whose encouragement played a crucial role in this achievement. This milestone reflects the collaborative spirit of our club, and we hope it serves as an inspiration for future members to approach new challenges with the same dedication and enthusiasm. Thank you for your trust and support in making this journey a success.

Warm regards,

Sanjana R & Harsha B

Stage Committee IC Design Club – Deccterous Department of ECE,

NIE Mysore

Dear Members,

As the Idea Lab Projects Committee of the IC Design Club- Deccterous, we are pleased to acknowledge the successful completion of this year's newsletter, which reflects our collective efforts and shared vision. We extend our heartfelt gratitude to everyone who played a part in the successful launch of this newsletter. Your efforts, dedication, and support have been instrumental in bringing this publication to life, and we deeply appreciate your contributions. A special note of appreciation goes to the Editorial Team for their dedication and hard work in putting this newsletter together. Their efforts have resulted in a publication that truly represents our committee's achievements. Our collaboration with the Idea Lab has played a key role in fostering innovation and providing a platform for students to explore and develop their ideas. It has encouraged creative problem-solving and interdisciplinary learning, and we look forward to continuing this association while supporting impactful projects. We are sincerely grateful to our faculty advisors for their constant guidance and support. Their encouragement has been instrumental in shaping our journey, and we truly appreciate their invaluable mentorship. Thank you for the trust and the opportunity to contribute to this incredible journey. We look forward to continued growth and success together.

Warm Regards,

Shraddha Satish Amalkar ,Shivabasavesh A S

Mourya P

Idea Lab Projects Committee IC Design Club-Deccterous

Acknowledgment from other committees

Dear Members,

As part of the Technical Committee of IC Design Club – Deccterous, we take great pride in recognizing the collective effort that led to the successful completion of the 2024 Club Newsletter. This accomplishment is a true reflection of our members' passion, dedication, and teamwork in bringing a compelling and insightful publication to life. Our vision was to craft a newsletter that not only highlights the club's achievements but also serves as a dynamic platform to showcase technical innovations, thought-provoking ideas, and creative contributions. Through meticulous planning, technical precision, and seamless execution, we have created a publication that embodies the spirit and ambition of our club. We extend our sincere appreciation to the Editorial Team, whose expertise in curating and refining content has given this newsletter depth and impact. A special mention to the Design Team, whose creativity and attention to detail have made it visually striking and engaging. Additionally, we are grateful to our faculty advisors for their unwavering support and invaluable guidance, which have been instrumental in shaping this initiative. This achievement stands as a testament to our club's ability to collaborate, innovate, and excel. We hope it inspires future members to take on new challenges with confidence and enthusiasm. Thank you for believing in us and for allowing us to contribute to this remarkable initiative. We look forward to continuing our journey of innovation and excellence.

Best regards,

Jayesh Gidwani & Nidhishree

Technical Committee IC Design Club Deccterous Department of ECE, NIE

Dear Members,

As the Publicity and Sponsorship Committee of the IC Design Club - Deccterous, we take immense pride in acknowledging the successful execution of our initiatives this year. Our combined efforts have strengthened the club's outreach and reinforced its presence. We extend our sincere gratitude to everyone who contributed to making this possible. Your dedication and enthusiasm have played a crucial role in enhancing the visibility of our club and securing valuable sponsorships. A special thanks to our dynamic team for their commitment to ensuring effective promotion and support. Our collaboration with various sponsors and partners has been instrumental in providing the necessary resources to fuel innovation within the club. We deeply appreciate their trust and support, and we look forward to building even stronger associations in the future. We are especially grateful to our faculty coordinators, whose unwavering support and guidance have been invaluable. Their efforts in facilitating our work and ensuring smooth coordination have significantly contributed to our success. A special note of appreciation goes to the editorial team for their hard work in curating and presenting our initiatives in an impactful manner. Their dedication has resulted in well crafted content that highlights the club's achievements and outreach efforts. Thank you for the opportunity to be a part of this incredible experience. We look forward to achieving new milestones together.

Warm Regards,

SKANDAN BHARADWAJ K & SATYADUTT

Publicity and Sponsorship Committee IC Design Club - Deccterous

Acknowledgment from other committees

Dear Members,

As the Project and Industrial Visit committee of the IC Design Club- Deccterous, we are happy to extend our heartfelt gratitude for the successful completion of this year's newsletter. It was a tough but rewarding journey of organizing the field visit and coordinating with other committee members as well. The visit to Lahari Electronics was a resounding success, and your active involvement played a crucial role in making it a memorable experience for all involved. The insights gained from this visit, as well as the opportunity to interact with industry experts at Lahari Electronics, have undoubtedly broadened our knowledge and provided a deeper understanding of the IC design and electronics industry. We extend our unwavering gratitude to the editorial team for their efforts in ensuring their seamless coordination and well curated contents throughout the year. A special thanks to the hospitality committee and sponsorship committee for their pivotal role in organizing our field visit. We would like to thank our faculty advisors who invested a lot of their time and efforts throughout the year. Their constant encouragement and involvement have been instrumental in shaping this initiative. We are grateful to be associated to this club and the opportunities it has to offer. We look forward to be part of this club as it reaches new heights.

Warm Regards,

Adrutha ,Pranav, Bheema ,Vishanth

Project and Industrial Visit committee IC Design Club - Deccterous

Dear Members,

As the Hospitality Committee of the IC Design Club - Deccterous, we are incredibly proud to acknowledge the successful completion of our 2024 Club Newsletter. It is a privilege to be part of a team dedicated to creating a welcoming and enjoyable experience for all attendees We sincerely appreciate the opportunity to contribute to the committee's mission of ensuring a warm, welcoming, and seamless experience for all guests and participants. The responsibilities that come with this position, including guest coordination, event planning, or hospitality services. We are committed to contributing my time, effort, and enthusiasm to ensure the success of our initiatives. Thank you for this opportunity to serve. We look forward to working alongside my fellow committee members and making a positive impact.

Warm regards,

Varshini H R,Hemashree G N, Neha S

The Hospitality Committee of IC Design Club - Deccterous

Department of ECE, NIE Mysore

Acknowledgment from Editor

Dear Members,
I am extremely delighted to express my heartfelt appreciation for the opportunity to edit and present the 2024 Club Newsletter. It has been an incredibly cheerful and fulfilling experience to compile and showcase all the remarkable achievements our club has accomplished over the past year.

This newsletter stands as a testament to the dedication, innovation, and collaborative spirit of our members. Every milestone, technical breakthrough, and creative contribution featured in this edition reflects the hard work and passion that drive our IC Design Club – Decctrous forward.

A special thanks to our editorial team, contributors, faculty advisors, and mentors for their unwavering support and guidance throughout this journey. Your contributions have played a crucial role in making this publication a success.

Presenting this newsletter fills me with immense pride, and I sincerely hope it serves as both an inspiration and a record of our collective excellence. Let's continue to strive for greater heights and celebrate many more achievements together.

Thank you all once again for your support and enthusiasm!
Warm regards,
Anagha M K
Social Media Committee
IC Design Club – Decctrous,
Department of ECE, NIE Mysore

Contact section

Stay Connected

For updates, events, and more, connect with us through the following channels:

Website: https://icdesignclub.ccbp.tech/\

Instagram id: DECCTEROUS.IC_DESIGN_CLUB

LinkedIn Handle: https://www.linkedin.com/company/deccterous-nie/

For any inquiries, please feel free to contact via email or through LinkedIn.

We appreciate your continued support and look forward to stay in touch.

Best regards,

DECCTEROUS

Ideate • Innovate • Create • Perpetuate